

Concorde Heritage avionics package upgrade



Concorde Heritage avionics package upgrade 2.0

Developers notes and manual

for Colimata Concorde FXP 2.0

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Foreword

This manual include procedures how to operate some specific instruments which are not documented or present in original Concorde FXP package.

When i was bought Colimata FXP version 1.00 years back i was not happy with lack of functions and incomplete visuals. With my little knowledge of LUA programming and basic 2D/3D modeling i was able with help of community users bring this awesome modification full of new features to you, for free. But donations are welcome.

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Chapter I.

Here is list of what is new or different from original Concorde FXP:

Fully operational CG computer with all displays and knobs

Fuel consumed counters now operational

Engine control schedule now simulated and affect N2 and EGT instruments

Improved engine instruments behavior

Plessey PVS1580 data acquisition computer, first time ever simulated

Visual windshield icing

Three independent chronometers all with timer function

Automatic fuel trim transfer with load limiters

Takeoff CG 54% switch operational

Improved Decision height alert

ISA temperature gauge now simulated on FE panel too

Altitude, temperature and speed indicators new in FE panel

Improved exterior lightning and cabin ambient lightning

Improved cabin temperature settings

All gauges are now working at Engineer's panel. Every system is now modelled, some fire, icing and undercarriage tests are not.

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Cabin pressurization is now in manual mode, you have to set desired cabin altitude and cabin rate, system will pressurize the cabin.

Oxygen system for cca 20 minutes is available from breathing masks in case of emergency from hypoxia or blackout.

Fuel management with CG adjustments is at your hands now, fuel pumps are operational and transferring fuel within tanks.

Refuel system for stop over are available and in same manner the jettison fuel system.

Automatic fuel trim settings are set by default, no need to go to main menu. Just keep the CG within the limits and will be set automatically.

Intake system can be selected by manual or automatic, however in manual mode changing the ramps is still not possible.

Engine status lights now react to overheat, failures, start and wind down.

Fuel temperature is now simulated, regulate it by heaters.

Emergency relight simulated for relighting failed engine when airborne.

RAM air turbine and 5th emergency generator available in multi engine failure or power loss.

Aircondition tests available, indicated by corresponding warning lights.

Fuel panel has now many status lights like overfull, low pressure and low level, all reacts to fuel quantity.

Hydraulic and electric panels are the key for sufficient power to all systems, both have now proper logic with backup systems, keep them under pressure.

All engineer's panel lights have tests switches, actually three, if you find some which dont lit, that means these have no function now.

Master warning sounds can be muted now with the correct switch on back of the engineer's panel. Also volume of it can be changed.

Use loudspeaker switch to switch between default and Heritage warning system.

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Various overheat, fire and smoke tests from engineer's panel.

Autopilot modes for go-around and backbeam still present.

These are now ready to use as very helpfull autopilot modes, look for flying manual of the Concorde.

We have warning lights and sounds in flightdeck:

The logic for various switches and scenarios has been added including warning lights with sounds so

now you can easily find what is working and what is not working in the Concorde.

Enhanced starting procedure:

Now you really need to do it right, because it wont let you start the engines if you have no ground power operational.

Follow the Concorde checklist if you cannot power it up.

Braking system:

Brakes can overheat now, use fan to cool them down and prevent tire blow in worse scenario. Associated warning lights works now.

Brake tests are now working with temperature gauge too.

Default positions of some switches including reheats has been changed. Fix for reheats has been removed.

Radiation detector now "works":

Yes, real Concorde has Geiger-Muller tube for radiation detection. As it flies much higher than other airliners radiation level is monitored.

There is working gauge for radiation level. I have simulated radiation level by random number generator in nonlinear scale so the risk is very low and you may not see the red warning ever. However if you will see it you need according abnormal procedures descend to lower altitude.

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Icing conditions warnings:

As part of warning lights the ice warning lights react to x-plane default icing model.

Because default X-plane icing model does not work on ground i recommend using my other addon Icing simulator which is more real and works well with Concorde FXP. Now you can get visual icing on windows.

Autopilot enhanced:

It adds autopilot function for selecting direction of the flight so you dont need to use controller (joystick/VR) to turn the airplane left or right.

It overrides joystick turns if general AP1 is ON. If you turn AP1 off you are back with hands on your controller or joystick.

Use black AP fine tune knob turning easily left or right and Concorde will start turning. There is also bank limit 30 degrees.

Separate power on engines:

Another feature is in separate file 4heavy.lua and can be used with all aircrafts regardless* of Concorde but handy with Concorde.

It allows you to use only 2 of four engines on each side of wing while turns or so. By default if you have dual throttle controller

all 4 engines are stick together, this will separate them for two pairs. 2 on left 2 on right.
Compatibility with 2 or one engine

powered planes is maintained. Extra/modified profile in X-plane 11 controller config needed.
Instructions below. Do not work with

single or quad throttle axis controllers and in VR, only for dual throttle.

*-used by profile for 4 engines planes, other planes unaffected

Correct V/S selector:

Another feature enhance vertical speed hold.

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Use it like in real airplane by the up/down selector next to the AP turn knob.

Heritage sounds and radio mode:

Next feature are sounds between tower and BA Concorde during last flight. Many of these sounds from

JFK to LHR are played automatically. Only clearance, takeoff and one special sound you can call whenever you want.

Those historic ATC sounds brings more authenticity to this famous plane.

For IVAO users you can now benefit from 8.33kHz model on COM1/COM2 with the patch of the radio, also frequency range is now corrected 118 to 137 MHz.

Outside Lights patch:

Next patch is modification for outside lightning. It will add spectacular effect for landing and taxi lights. Also contains correct strobe lights.

Last patch is correct cockpit and FE panel colors of the warning lights, original release have mostly red.

Section 1. Power up

First time in pilot seat? Either yes or no some good practices here. Follow checklist provided by Colimata for other steps, set the route plan and adjust autopilot etc. Below are only specific procedures which differs or are not documented.

Now is good time for checking the switches if they are at off positions. Some fuel pumps could be ON turn them off to prevent unwanted operation.

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To have power, you need ground cart truck with 115V. Call it with yellow GRND Call push button at overhead panel plus on Engineer panel bottom left part here is next to Ram air turbine switches Microphone selector, switch it to ground. Now you will see at Engineers panel right part electricity panel that GROUND POWER light is on.

Dont forget turn ON batteries, A and B on right engineer electricity panel. Last action is two switches must be ON in this panel. Ground power switch to close and SSB to close. SSB switch is just below the Ground power switch. Now all is powered.



Section 2. CG computer settings

Before engines start Center of gravity computer should be set. In the center of Flight engineers panel is CG unit with computer and weight selectors.

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Label with knob show ZFW. This is loaded aircraft weight without fuel. If you don't know this value, you can get it from Plessey data computer on bottom right Flight engineer panel. Details in Plessey PVS1580 section. After that turn the knob and preset value for ZFW. To lock it push the button. ZFCG is similar but it is currently predefined inside x-plane and entering the value has no effect. A/C Weight display needs to be set. Right knob is at N mark. Turn it counterclockwise to each dot from N and set ZFW. At every dot push and hold for one second left hand knob until number change.



Careful here, ZFW for this display is 10x lower than real ZFW, for example 88330kg you enter like 08833. When you reach 5th and last dot, go back to 4th dot, this will start setting FUEL REM display which is actual fuel on board, again 10x less than original, so 94123kg is 9412. Push again left hand button like before at every remaining dots clockwise, when you get back to N the fuel will be added to the A/C weight and from now any fuel burned will be deducted from these displays showing weight and fuel decrease.

If you push and hold at N left hand knob for more than 3 seconds it will erase all readings and you can start again if you made mistake or not working.

Section 3. Fuel trim for takeoff

After engines started, you will hear call out "trim for takeoff" announced by virtual flight engineer. If not, that means CG is in front and must be adjusted.

You don't want to ask passengers to take another seats you would rather move the fuel aft. To do it you need to operate some fuel pumps and inlet valves.

You can do it manually or use automatics. Manually open inlet valves on tank 11 and turn ON pumps from tank 9 if it has fuel. This will move the fuel aft.



To use automatics change lower/raise load limits and use Trim auto master AFT switch under red cover, it will do the same by one switch.



Colimata implemented for easy use Trim callouts/settings for beginners. You can set it from menu or leave it for automatics which is watching current CG position as fuel is moving and other engine rating settings. Here is the sequence and requirements.

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Takeoff - CG must be 52.7+ and Engine rating switches at overhead panel at Takeoff

Flight - CG must be 53.5+ and Engine rating switches at overhead panel at Flight, if they are not at 18.000ft serious stability issues occurs. Can be overridden by menu preset

Descend - not defined

Landing - CG must be below 55% and Engine rating switches at overhead panel back at Takeoff.

Ground - CG below 52.7% and ground speed below 100 knots



The automatics work only if at FE fuel panel is ON switch FUEL LP Bypass. Otherwise is deactivated and you have to do it manually. FUEL LP Bypass is mandatory if you want to use manual trim settings from menu. Not recommended and not optimized in Heritage Package this manual preset mode. Many new features are disabled in manual mode from GUI.

Section 4 - Before takeoff drills

Before takeoff there are some important procedures which you will not find in standard Colimata FXP.

One critical is preset Cabin pressure altitude and Cabin rate. You can do it after takeoff but there is too much work and you can forget it.

Top left side of FE panel there are System 1 and 2 gauges. Use one of your choice to select desired cabin altitude for pressurization system. Usually 6.000ft with knob marked A. Don't forget to select cabin rate with R. 180° turn clockwise is about 400 feet per minute. The indicator will change after takeoff and it takes some time to reach preset. At about FL150 you should get the Cabin rate automatically steady at 0.

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If not do not climb otherwise you may have black out or darkness because Cabin is not fully pressurized. In emergency you can wear oxygen mask, take one of three in the cockpit. Before you can use them, black rotary selector at bottom right FE panel must be turned and crew gauge of oxygen pressure must read at about 70 PSI.



Second is setting Takeoff monitor. To ensure all engines provide enough thrust for takeoff calculated fuel flow and pressure must be set first otherwise the green to go lights on primary engine instruments will not light up and takeoff is not recommended under these conditions. To set minimal fuel flow look for primary engine instruments fuel gauge and there is small rotating knob for each engine and turn it to desired value. The values are currently two preset minimal for takeoff with reheat around 22tons/h and takeoff without reheat 11tons/h. P7 or pressure preset knob is at FE panel left side on secondary engine instruments. Again now is static presets which for takeoff with reheats are 39 PSI and without reheats around 16 PSI maybe a bit more. After that settings for all engines push Engine monitor to engage during takeoff and it will show or not show green lights on primary engine instruments.

Section 5 - Fuel Balance during climb and cruise

In section 3. was required CG marks to prevent stability problems. To achieve them fuel transfer is necessary. Simply to move fuel to the aft use tank 9 and 10 and transfer fuel from them into tanks 5 and 7 or tank 11. Use tanks 5 and 7 to refill constantly tanks 1,2,3 and 4 connected to the engines. This technique is the same as in original Concorde FXP. What is different is aft fuel trim with load limiters presets in center FE fuel panel. This switch under cover when uncovered gives automatic forward move or rearward move fuel. If the load limiters counters are set to certain value, it will leave that preset value on the respected tanks to maintain future trimming because trimming is possible only between tanks 9 and 11, very limited transfer in other tanks. If you for some reason drain all trim tanks 9,10,11 last chance to shift fuel is to use trim tanks switch 1&4 which if tanks 1 and 4 are not refilled and level goes below 1.7 tons each will when refilled maintain this level results a small aft CG move.



Please note several pumps have auto setting, this is set by this auto trim master switch, if not used switch them to off because AUTO may be considered as ON. Same as inlet valves.

Do not misunderstood standby inlet valves, they are in line and in Heritage package works only for refuel at ground for now.

In Original Colimata FXP they dont have refuel capabilities and they are used for abnormal fuel transfers between all tanks. As much as possible have all tanks 1,2,3 and 4 filled because if one reach empty, associated engine will shut off.

Section 6 - ECS Engine control schedule and ISA temperatures.

Engine control schedule now controls N1, N2 and EGT parameters of engines during various phases of flight for fuel efficiency and noise limits.



The only important note is when you go over Mach 1 do not leave ECS at Flyover settings, causes warnings. Keep in mind that Engine instruments now react to OAT and ISA temperatures, in extreme conditions you may have engine instruments warnings and throttles must be adjusted to meet safety limits. If they are exceeded, they will be logged into PVS1580 data recorder for further evaluation.

Chapter II.

Here are explained some differences you will not find in original Concorde FXP for Pilots and Copilots area.

Section 1 - Audio warnings and Heritage sounds

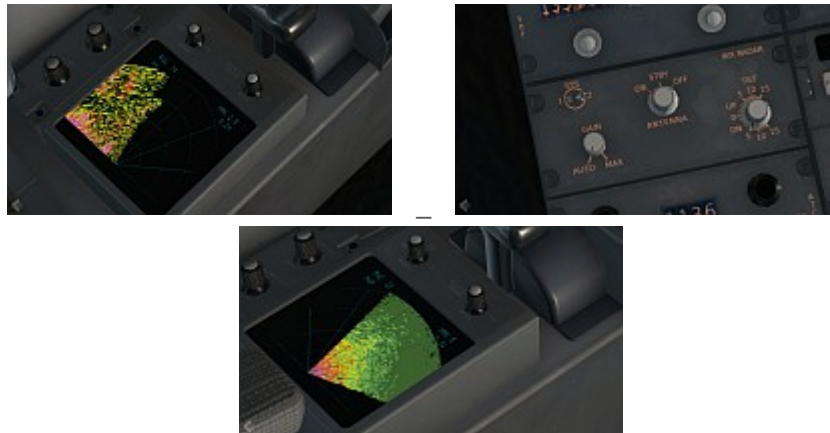
Many of pilots and copilots audio knobs and switches has volumes of different callouts from last Concorde flight as a tribute and a little fresh ambient experience when the flight become quiet and boring. By default the sounds are at low volume, to get all them available turn volumes up on center console. Some sounds will come up automatically during flight. Some are triggered with green push buttons or TO monitor switch during taxi. If you want more audio warnings look for loudspeaker switch at pilots left arm.



Section 2 - Weather radar

From last Heritage package version is available Weather radar, it has also capability of ground proximity system display but not warning.

Only one display is available showing same data on both units. Turn them on with ON/OFF/TEST switch. You can change the range on Pilots unit between 10 and 40 miles, to get greater view use Copilots left knob when Pilots is at 40, it will reach upto 160 miles. For weather display shows WXR, or WXA where A means visual alert of bad weather.



To have another alert use copilots right arm located Radar alert switch, when bad weather will be in vicinity white flashing light appears near autopilot area. To have Ground proximity radar on screen look for center console where are other switches for WX radar. Switch SYS1 is WX radar, SYS2 is Ground proximity radar. You can adjust the focus by antenna angle up and down and gain too. You can turn off radars by on/standby switch from here too. Radar needs ground power or engines running to operate.

Section 3 - DTG counter and Timer

On center console is green LED area used for Passengers information displays located in Cabins. It is not now connected but you can use it for your purposes measuring remaining distance to destination. Turn the black knob to preset distance to go in miles divided by 10 so shown 300 will be 3000nm because there is painted zero. When you switch it on by left switch next to the display after takeoff will start decrease by flown miles. It can be reset only after it reaches zero and must be preset more times until it shows correct value.



Timer is new capability of all three now independent clocks. Third is at top left FE panel. Elapsed time setting is available only at Pilots seat. Chronometer for all and timer too. To use timer switch the main switch from chrono to timer, then top switch from hold to left positions where first is seconds and last is minutes. If you wait at one of positions it will start increase time automatically. Back to run will start countdown. To reset it go left to reset.

Section 4 - Enhanced autopilot modes

An go-around mode is available here. It is activated when you are on landing phase below 1000ft and you need to abort landing. Dont panic, just push throttles to full and go-around mode will be activated automatically. It will maintain pitch 15 degrees until 1000ft is achieved, then will be disconnected automatically. You can now turn on backbeam mode if you were on localizer to follow it after go-around, limit is 30 miles.

Vertical speed can be set on center console at autopilot control, exactly pitch datum. FD1 must be on first to see selection in VS indicator. With AP1 on VS autopilot modes during flight it will follow preset vertical speed.

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Turn knob on center console can be also used as primary lateral navigation if no other mode is set, use this black rotating knob to turn left or right.

Bank limit is 30 in this mode. For some reason it cannot be used with MAX CLIMB or MAX CRUISE.

Section 5 - Improved Decision height and waypoint change warnings

There is only one small improvement, if you preset during landing DH altitude by red knob you may hear sound warning when DH is reached.

Waypoint change alert will show flashing yellow warning near autopilot area when next waypoint is within one minute.

Section 6 - Icing

We implemented visual icing on windshield and side windows. So you can now see building of ice during cold weather.

Both Visor and windshield are modeled. To get rid of ice simply turn on windshield de-ice at overhead panel.



Another ICE warnings and de-ice systems at overhead panel are connected to x-plane icing system. I strongly recommend use of my Icing Simulator v1.20 available on x-plane forum for better icing experience because default x-plane model is poor.

Section 7 - Air data computers ADC

On center console are ADC 1 and ADC 2 computers comparing various sensors and instruments, if there is discrepancy now pilot will be warned. Both have some TEST modes, the longest takes 35 seconds and every test is completed when blue TEST light is on. Most of the tests are available only at ground.



Chapter III.

Flight Engineer's panel was heavily modified before Concorde FXP 2.0 and after that it still have extra features, here are some differences explained.

Section 1 - Brakes gauge



Brake gauges has temperature area and pressure area. Pressure area reacts to pressure applied to brake pedals. Located at FE panel top left.

Temperature gauge shows actual brake temperature and if one of the wheels warning is pushed it shows current wheel temperature. There is test button for gauge test. If you hit more than 700°C wheels may got puncture from heat.

Section 2 - Intake pressure error gauges



Below temperature brake gauge are four indicators of intake pressure error. This now works from Mach 1.3 and shows pressure difference error between intake moving ramps. If the pressure difference is bad needle lean left or right to the yellow bands. You should adjust throttle settings if so.

Section 3 - In flight engine restart



If for some reason one or more engines fail during flight and there is no permanent damage on it, you may try restart them. To do this you need start emergency generator on right FE electricity panel. Unguard the switch and turn it on, blue light will come up. Then use rotating knob called relight which is bottom left FE panel near engine start switches. Turn it from off position to affected engine number and try to start it like normally. After successful restart turn it back to off.

Section 4 - Additional pressurization switches.

In Chapter I Section 4 was described basic functions of pressurization system. Additional information needs to be told.

Before landing cabin must be depressurized. To do this turn the Cabin altitude A knob down to zero, completely to zero and R rate knob counterclockwise to the limit or until the rate will go to negative values. Do this only below FL150.

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System select switches maintain in pairs, SYS2 is backup system. Discharge valves for respected systems are normally at NORM position but may be close the valves with visual indication now. Ditching and Emergency depress are for emergency situations and works, Ditching is prevention of flooding interior when emergency landed on water and depress is used as rapid 4000ft per minute depressurization. Excess alt and overpressure warnings are shown when system operates beyond limits.

Section 5 - Additional information for Engine control schedule ECS



There are 4 lights showing performance, LO is default maximum power used for takeoff and reheated climb. HI is the opposite adjusting N1/N2 ratio for maximum fuel efficiency used during cruise. MID is for approach and FlyOver is for noise abatement procedures flying above land with less than 95% of power.

Section 6 - Fuel tank pressure gauge and Radiation gauge

At lower left FE panel you will find Fuel Tank pressure gauge. This read normally 0 but after FL400 it is automatically pressurized to about 1.5 PSI. If you climb too fast can reach more than this until warning shows overpressure. In this case slow down climb to maintain 1.5 PSI. Same situation during descend.

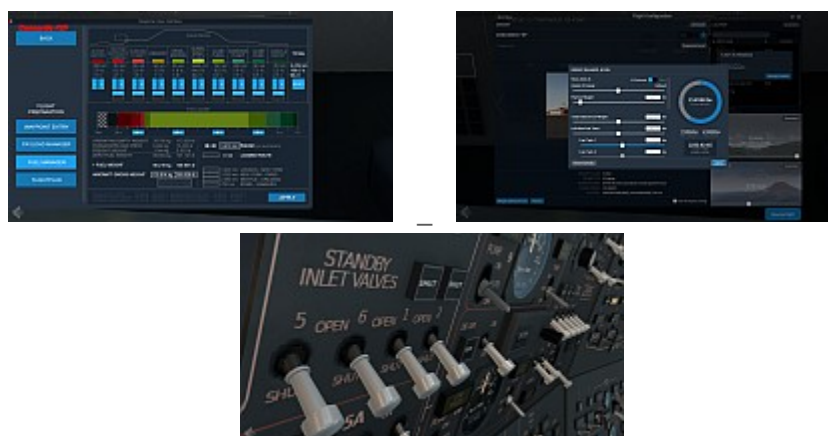


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Below is one extra instrument, reading milirems as Radiation counter. This is randomly generated to simulate solar and ambient radiation. At FL600 and closer to earth poles the radiation is highest.

Section 7 - Refueling, CG computer modes and Fuel consumed displays

Fuel should be loaded before powering the aircraft. You can use Colimata GUI or simply default x-plane weight and balance menu where you can only estimate the amount by percentage of roller bar because the units are wrong. There is possible to do it after powering aircraft but you have to apply refueling procedure specific for Heritage package. To do it you must be on ground and set all inline standby inlet valves in middle of FE fuel panel to ON, all of them. After that you will see that Fuel total contents differs from Captains/Copilots total fuel contents counter and fuel in tanks starts increasing. Normally it should stop refueling when the limit is reached.



If will not, stop it manually by closing that standby inlet valves. This procedure is used normally for refueling after stop over.

CG computer has on top several test modes. One is Gauge tests, it will test needles moving. Next are several Min Max tests, this is only internal comparator tests, will show on displays comparator warnings. 1 and 2 CG tests are again internal comparator tests.



Filament tests shows 88.8 at CG display. After tests move the selector to OFF position and TEST switch to off too.

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Master CG mode selector is normally at M. If some gauges stops reading or reads wrong values you may set it to 1 if failed gauge is on right side and select 2 if failed gauge is on right fuel panel. These modes 1 and 2 use mirrored copy of gauges. Does not affect trim tanks 9,10 and 11).

Fuel consumed displays show used fuel counter and have one knob for test and reset. To reset it, turn it to one side, to test it to another one, the knob will turn back automatically. The orange LEDs are fuel density compensation error indications.



Fuel density is not simulated so it is completely random. Fuel dump or Jettison fuel can be done but is not recommended because Colimata put very high rate and you may lose too much fuel within one minute.

Section 8 - Plessey PVS1580 data acquisition computer

This is the first time ever simulated Air data computer system in Concorde. It is used for monitoring and diagnostics of performance data of engines, speeds and so on. It uses 4 digit codes which everyone means one performance data. List of codes are in Main GUI menu under custom content.

To use it on ground, as it is usually done after landing switch the main switch to ON. If you want to see real data during flight it will be switched by default to POWER or after disconnecting the ground power. Use keypad to enter code, when all 4 digits on left are displayed, hit DIS key to show on right the data. If no data is available it will show zero. To enter new code hit any key to data entry mode and put the correct code now. If you want to clear the data display use CLEAR key. After stop over or before next flight CRL MEM clears all memory.



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Rotary selector selects corresponding unit with recording tapes to red fail light. SYS is general mode for all units. However only AQ1 is without fail light because simply no more tapes are available today and old PDP 11 readers used for data collection are in museums.

If you see Green light that means some limits have been reached and on code display you may see which one or which category it is. If you hit DIS in this mode you will see the percentage exceeded that limit.

Section 9 - Oxygen system gauges



At the bottom right FE panel near the floor there are gauges of pressure for passengers and flight deck. After takeoff turn the black knob to open the oxygen system for the oxygen mask in case of emergency in case of black out. If you wear a mask for long time the pressure will slowly decrease as the oxygen is breathed in. The oxygen bottles capacity is to about 22 minutes. It can be refilled together with refuel procedure.

Chapter IV. - Miscelanous features

ISA temperature is normally not modeled in the X-plane today. Because temperature is key factor at Mach 2 many temperatures now affects performance and one of them is ISA which is International standard atmosphere. If it shows +10 it means the day is hotter then standard and on hotter air engines produce less power. This altogether means for you to carefully watch the temperature gauges for engines and Total temperature indicator which is 127°C.

Various tests of warning lights, fire tests, aircondition tests, smoke tests are available, but not all are working. This should be done before every flight. Dont forget turn the selectors back to off before flight.



In the fuel system is automatic scavenge tank and pump operating. If there are too much fuel flow or pumps running at same time the excess fuel is stored in this hidden tank. After the TANK 3 is not full, all fuel from this scavenge tank will be slowly transferred to TANK 3.

Chapter V. - Notes

X-Plane has failures mode available, if you have this set on by default you may with Heritage package occasionally seen some systems in failure because i made it connected. Because x-plane generated failures are usually too aggressive in this case look for nearest airport and try to land or simply disable the failures and fix them by one click to continue. Some systems may fail even when the failures are off because you may run over limits.

I said that ECS and temperatures affects performance, but only at cockpit gauges, real x-plane model is untouched in this version.

This means you still have 100% power by x-plane model even if gauges shows you are over limits or below limits. This is for compatibility for now.

If you did not understand how the systems work, use checklists or watch my tutorial videos on youtube.

Happy flying!

Version 2.00

Gaston 2022

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